

LEV SELECTOR, Ph.D.

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- <https://www.linkedin.com/in/levselector>

- <https://www.youtube.com/@lev-selector/videos> - AI, 6K+ subs, 270+ vids

- <https://github.com/lselector> - many lectures, tutorials, code examples

- <https://www.amazon.com/s?k=Lev+Selector&i=digital-text> - 4 AI books



SUMMARY:

- Artificial Intelligence AI, Large Language Models LLM, Agents, MCP, Skills, RAG, Vector DB, AI Apps
- Machine Learning, Data Science, Analytics, Expert, Leadership, Team Builder
- Ph.D. in mathematical modeling and computer simulations
- Experience of building teams of engineers and data scientists (hiring, training, and managing)
- 15+ years of experience with high-volume data processing, analytics, and modeling
- Proven track record of building data and analytics systems from scratch, and delivering them on time
- Full cycle development from requirements and planning to architecture & implementation
- Experience in different areas - Financial, Advertising, e-Commerce, Media, Publishing, Crypto
- Data Engineering on clouds (AWS, Azure, GCP) and on-prem, ETL, APIs, big data, data mining
- Mathematics and Finance: Ph.D. (Math. Modeling), Advanced Calculus, Vectors & Tensors, Theory of Complex Variables, Differential Equations, Differential Equations in Mathematical Physics, Probability & Statistics, Statistical Radiophysics, Optimization, Monte Carlo Simulations, Numerical Methods, Machine Learning, Deep Learning, NLP, Transformers, Generative AI, LLMs
- Programming & Data Science Tools: AI Coding Assistants (Cline, Claude Code), MCP, Skills, Agents, Python, Pandas, NumPy, Rust, SQL, ETL, FastHTML, Ollama, LangChain, LangGraph, Scikit-Learn, NLP/NLTK, Huggingface tools, AWS SageMaker, Perl, Go (Golang), Javascript, C/C++, Java, Excel VBA, SQL databases (PostgreSQL, MySQL, Vertica, Netezza, Google BigQuery, MongoDB, Sybase, DB2, Oracle, MS SQL), Web Apps, Cloud (Amazon AWS, Azure, Google, IBM)

PROFESSIONAL EXPERIENCE:

Oct 2024 - present - Vareto LLC (<https://vareto.com>) - AI Development, AI Architecture for SaaS

- AI, Vibe Coding, Cline, Claude Code, Claude, Gemini, AI Agent for SaaS app, Rust, Python
- AI-enabled ETL, Excel files parsing using AI

Feb 2024 - Sep 2024 - RSM US (<https://rsmus.com>) - AI for audit

- AI, LLM, Vibe Coding, Audit, Documentation, RAG, AI to generate audit workflows and Visio diagrams
- Python, OpenAI API, Huggingface tools, FastHTML, LangChain, Statistics, Machine Learning,

Nov 2022 - Sep 2023 - CTO & Co-Founder, BixBeta

- AI-enabled crypto accounting and payment/donation
- Generative AI to add Natural Language Interface to analytics and reporting

Nov 2020 - 2022 - Data Science Practice Lead, Redapt

- Building, training, and managing a team of data scientists
- Data architecture in the cloud (Azure, AWS, Google), Apache AirFlow, ADF, SQL Data Warehouse
- Hands on Machine Learning modeling in the cloud (Azure Machine Learning Studio, Azure Synapse Analytics, Azure Cognitive Search, Python, Scikit-Learn, AutoML, predictive modeling, regression, classification, time series forecasting, anomaly detection)

Feb 2020 - Nov 2020 - Selectorweb Inc., consulting projects

- National Debt Relief (<https://www.nationaldebtrelease.com>) Machine Learning and ETL in production
- Provisions Group (<https://provisionsgroup.com>) Machine Learning with AWS SageMaker and R
- Vara (now <https://vareto.com>) Data Integration for analytics

Dec 2019 – Feb 2020 - Head of Machine Learning And AI, Digital Labs, Capco

Built, trained and managed a team of data scientists, hands on modeling using Python, AWS, Linux VMs, Mainframe Test Automation project using Python, GnuCOBOL, libFuzzer, antlr4, Neo4j

Dec 2018 – Sep 2019 - VP of Data Science, Head of Analytics - National Debt Relief, LLC

Built, trained and managed a team of data scientists; created architecture on Linux server(s) on Azure cloud to run hundreds of jobs (ETL, data processing, Machine Learning); building visualizations using Salesforce Einstein Analytics Dashboards and Jupyter notebooks. Azure SQL Data Warehouse, Python, Nginx web server, Unicorn gateway server, Scikit-learn, Random Forest, XGBoost, H2O.ai)

Jun 2018 – Dec 2018 – Senior Data Scientist at Vanguard Group via Tata Consulting Services

Machine Learning and Artificial Intelligence, Scikit-Learn, AWS SageMaker. Designed architecture for data migration into the cloud (AWS) for analytics, developed data structures and lambda functions. Managed deployment. Trained a team in ML and AI (hands-on and lectures). Built models for cybersecurity & anomaly detection.

Jun 2017 – Jun 2018 – Selectorweb, consulting projects (Clients: Galvanize, SaleZoom, JKCF)

Machine Learning, AI, Analytics, working with imbalanced data, NLP (Natural Language Processing), Python, Scikit-Learn, NLTK, Pandas, Numpy, TensorFlow, Amazon AWS (EC2, S3), Google Cloud (BigQuery, MySQL), IBM Cloud (IBM Watson – Natural Language Understanding for Sentiment Analysis, Tone Analyzer, Personality Insights).

Nov 2014 – Jun 2017 - Penguin Random House, Consultant – Trained and led the team to build systems in Business Intelligence/Analytics, Big Data Collection and Integration (ETL/ELT), Machine Learning and AI, Reporting. Python, pandas, Netezza, SQL, Redis, AWS, Linux. Created system for high-volume parallel web scraping at ~2 Mln pages/day. Organized "Deep Learning Book Club" to promote the use of Machine Learning and AI in-house; worked with several business groups to create various data feeds and tools to clean and ingest data; created Python framework to work with IBM Netezza DB; wrote analytics tools for price analysis and estimation.

Apr 2012 - Apr 2014 – AppNexus, Inc, Consultant, Financial Data Analytics - Designed and implemented a billing and reporting framework for processing trading data (~60TB/day) across 2000 clients with custom rules (43 thousand lines of python code, self-recoverable ETL processes, health-monitoring, auto-back-testing to validate both data and code). Designed business intelligence & analytics systems (multiple reports, graphs, analytics database - MySQL). Linux, Python, Pandas, Vertica, Mysql, Hadoop, Hive, Git. Reduced data extraction time for 30 Bln data rows from 3 hours to 7 minutes. Trained and managed a team of developers. Taught courses on python/pandas/database/analytics.

1994-2012 - multiple "Wall Street" consulting projects (16 years):

Goldman Sachs, Morgan Stanley, JPMorgan Chase, Citigroup, Merrill Lynch	Waterhouse Securities, CantorFitzgerald/Espeed, CSFB, HSBC, WorldQuant
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Data Processing and Analytics using Unix, Perl, SQL (Sybase, DB2, MySQL), data preparation, data load/migration, ETL, Web design (HTML, CSS, Javascript), C/C++, Java, Jython, Excel VBA

1991-1994 Columbia University. Staff Associate - Mathematical modeling of dynamics of organic molecules. Distributed calculations (unix, C/C++), analytics and graphics using IGOR software on Mac.

1981-1991 National Cardiology Research Center, Moscow, Russia. Researcher - Real time data acquisition and computer processing in neuro-physiological experiments. Pattern recognition, classification, computer simulations of nerve impulse generation and propagation along C-fibers. Partial differential equations, Hodgkin-Huxley model, Crank-Nicolson, modified Runge–Kutta methods. Hardware/software design of medical equipment.

EDUCATION:

- 1988 Ph.D. in Mathematics (modeling of nervous coding), Moscow Institute of Physics and Technology
- 1981 MS in Automation, Moscow Institute of Physics and Technology (MIPT), majoring in computers, electronics and biophysics, Diploma - computer simulation of nerve activity

LICENSES, CERTIFICATIONS, COURSES:

Machine Learning, Deep Learning, Generative AI, Google Cloud). Data Analysis with Python and Pandas. SEC Registered Representative (Series 7, Series 63). CQF (Certificate in Quantitative Finance). Advanced Object Oriented Perl. C++ for Quantitative Finance. Advanced Excel for Financial Applications. Java 2 (Sun)

REFERENCES:

.. http://levselector.com/LevSelector_recommendations.docx